

8/09/17

→ RANK OF A MATRIX.

A matrix is said to be of rank, " n " if

1. It has atleast one non-zero minor of order " n ".
2. All the minors of order higher than " n " are zero.

→ Elementary Transformation of a matrix:

→ Equivalent matrices

Two matrices A and B are said to be equivalent if one can be obtained from the other by a sequence of elementary transformations.

They have same rank.

→ Row Echlon form of a matrix:-

1. All non-zero rows are above all zero rows.
2. The leading coefficient of a non-zero row is always right to the leading coefficient of the row above.
3. All entries in a column below a leading coefficient are zero.

note: Leading coefficient is the first non-zero element of non-zero row.

- Non-zero row has atleast one non-zero element.
 - zero row has all zero elements.
- * Number of non-zero row in row echlon form of matrix is equal to the rank of the matrix.

eg:- Find rank of matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 2 & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

(This is now
echlon form)

$$\therefore \underline{\underline{r(A) = \rho(A) = 2}}$$

GRAUSS JORDAN METHOD (To find inverse)

Those elementary row transformation which reduce a given square matrix A to the identity matrix when applied to unit matrix give the inverse of A .

eg- $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$

Ans $\left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & -3 & 0 & 1 & 0 \\ -2 & -4 & -4 & 0 & 0 & 1 \end{array} \right]$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 + 2R_1$$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & -2 & 2 & -2 & 0 & 1 \end{array} \right]$$

$$R_1 \rightarrow 2R_1 - R_2$$

$$R_3 \rightarrow R_3 + R_2$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 12 & 3 & -1 & 0 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & 0 & -4 & -3 & 1 & 1 \end{array} \right]$$

$$R_1 \rightarrow R_1 + 3R_3$$

$$R_2 \rightarrow 2R_2 - 3R_3$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 0 & -6 & 2 & 3 \\ 0 & 4 & 0 & 7 & -1 & -3 \\ 0 & 0 & -4 & -3 & 1 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 3/2 \\ 0 & 1 & 0 & 7/4 & -1/4 & -3/4 \\ 0 & 0 & 1 & 3/4 & -1/4 & -1/4 \end{array} \right]$$

$$\therefore A^{-1} = \begin{bmatrix} -3 & 1 & 3/2 \\ 7/4 & -1/4 & -3/4 \\ 3/4 & -1/4 & -1/4 \end{bmatrix}$$

→ NORMAL FORM OF MATRIX (canonical form)

Every Non-zero matrix A , of rank = r can be reduced to the form :-

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} \text{ using elementary transformation is called normal form of the matrix } A.$$

eg:- For matrix :-

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix}$$

Find non-singular matrices P and Q such that PAQ is in the normal form. and hence find rank of matrix A .

$$A = I A I.$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 - C_1$$

$$C_3 \rightarrow C_3 - 2C_1$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow C_3 - C_2$$

$$\begin{bmatrix} (1 & 0) & (0) \\ (0 & 1) & (0) \\ (0 & 0) & (0) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} = P A Q.$$

$$\boxed{\eta(A) = 2}$$

where, $P = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}$

$$Q = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

eg:-

$$A = \begin{bmatrix} -1 & 2 & -1 & -2 \\ -2 & 5 & 3 & 0 \\ 1 & 0 & 1 & 10 \end{bmatrix}_{3 \times 4}$$

$$A = I A I$$

$$\begin{bmatrix} -1 & 2 & -1 & -2 \\ -2 & 5 & 3 & 0 \\ 1 & 0 & 1 & 10 \end{bmatrix}_{3 \times 4} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4}$$

$$R_2 \rightarrow R_2 + 2R_3$$

$$R_3 \rightarrow R_3 + R_1$$

$$\begin{bmatrix} -1 & 2 & -1 & -2 \\ 0 & 5 & 5 & 20 \\ 0 & 2 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C_2 \rightarrow C_2 + 2C_3$$

$$C_3 \rightarrow 2C_3 - C_4$$

$$C_4 \rightarrow C_4 - 2C_1$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 15 & -10 & 20 \\ 0 & 2 & -8 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\cancel{C_3 \rightarrow 2C_3 + C_4}$$

$$C_2 \rightarrow 4C_2 + C_3$$

$$C_3 \rightarrow C_3 + C_4$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 50 & 10 & 20 \\ 0 & 0 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 4 & 0 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

$$C_3 \rightarrow 2C_3 - C_4$$

$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 50 & 0 & 20 \\ 0 & 0 & -8 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & -2 & -2 \\ 0 & 4 & 0 & 0 \\ 0 & 6 & 4 & 0 \\ 0 & -1 & -1 & 1 \end{bmatrix}$$

05/09/17

Consistency of simultaneous linear equations

Consider "m" equations in "n" variables.

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_m$$

These equations can be written as :-

$$AX = B$$

where,

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Consider augmented matrix $[A : B]$

Case 1. If $\rho(A) = \rho([A : B]) = \text{no. of variables}$
then the system of equations is consistent and it has unique solution.

Case 2 If $\rho(A) = \rho([A : B]) < \text{no. of variables}$
then the system is consistent and has infinite solutions.

Case 3 If $f(A) \neq f([A:B])$

then system is inconsistent and it has no solution.

Now, consider homogenous system of equations.

$$[A]X = 0$$

then,

Case 1 $f(A) = \text{no. of variables}$

$\Rightarrow$ system is consistent and has unique solution (Trivial solution $x=0$)

Case 2 $f(A) < \text{no. of variables}$

$\Rightarrow$ system is consistent and has infinite solutions.

Case 3 In homogenous system no solution case does not exist.

eg:- Solve:-

$$2x + 3y + 4z = 11$$

$$x + 5y + 7z = 15$$

$$3x + 11y + 13z = 25$$

Ans.

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 5 & 7 \\ 3 & 11 & 13 \end{bmatrix}$$

$$B = \begin{bmatrix} 11 \\ 15 \\ 25 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Now, system of equation can be written
 $AX = B$

Now,

$$[A : B] = \begin{bmatrix} 2 & 3 & 4 & : & 11 \\ 1 & 5 & 7 & : & 15 \\ 3 & 11 & 13 & : & 25 \end{bmatrix}$$

Use only row operations to change into row echelon form.

$$R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 2 & 3 & 4 & : & 11 \\ 3 & 11 & 13 & : & 25 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$[A:B] = \begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & -7 & -10 & : & -19 \\ 0 & -4 & -8 & : & -20 \end{bmatrix}$$

$$R_3 \rightarrow -1/4 R_3$$

$$= \begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & -7 & -10 & : & -19 \\ 0 & 1 & 2 & : & 5 \end{bmatrix}$$

$$R_2 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & 1 & 2 & : & 5 \\ 0 & -7 & -10 & : & -19 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 7R_2$$

$$= \begin{bmatrix} 1 & 5 & 7 & : & 15 \\ 0 & 1 & 2 & : & 5 \\ 0 & 0 & 4 & : & 16 \end{bmatrix}$$

$$\therefore \rho(A) = \rho([A:B]) = 3 = \text{no. of variables}$$

$\Rightarrow$ system is consistent and has unique solution.

Now, we will form equations:-

$$x + 5y + 7z = 15 \quad \text{--- (1)}$$

$$y + 2z = 5 \quad \text{--- (2)}$$

$$4z = 16 \quad \text{--- (3)}$$

$$\therefore \boxed{z = 4} \quad (\text{from (3)})$$

Put in (2).

$$y + 2(4) = 5$$

$$\boxed{y = -3}$$

Put in (1).

$$x = 15 + 15 - 28$$

$$\boxed{x = 2}$$

eg:- solve:-

$$2x - 3y + 5z = 1$$

$$3x + y - z = 2$$

$$x + 4y - 6z = 1$$

Ans. system of equation can be written as:

$$AX = B.$$

where:

$$A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 1 & -1 \\ 1 & 4 & -6 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\text{Now, } [A : B] = \left[\begin{array}{ccc|c} 2 & -3 & 5 & 1 \\ 3 & 1 & -1 & 2 \\ 1 & 4 & -6 & 1 \end{array} \right]$$

$$R_1 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 4 & -6 & : & 1 \\ 3 & 1 & -1 & : & 2 \\ 2 & -3 & 5 & : & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$= \begin{bmatrix} 1 & 4 & -6 & : & 1 \\ 0 & -11 & 17 & : & -1 \\ 0 & -11 & 17 & : & -1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$= \begin{bmatrix} 1 & 4 & -6 & : & 1 \\ 0 & -11 & 17 & : & -1 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$$\rho(A) = 2.$$

$$\rho([A:B]) = 2.$$

$$\rho(A) = \rho([A:B]) = 2 < \text{no. of variables}$$

$\Rightarrow$ system is consistent and has infinite solutions.

Now, equations:-

$$x + 4y - 6z = 1 \quad \text{--- (1)}$$

$$-11y + 17z = -1 \quad \text{--- (2)}$$

Put $z = t$.

$$y = \frac{17t + 1}{11}$$

(From (2))

Put in (1),

$$x + 9\left(\frac{11x}{11}\right) - 66t = 11$$

$$11x + 66t + 9 - 66t = 11$$

$$11x = 2$$

$$x = \frac{2}{11}$$

eq. solve:

$$x + y + 2z + w = 5$$

$$2x + 3y + z + 2w = 2$$

$$4x + 5y + 3z = 7$$

Ans.

$$[A : B] = \begin{bmatrix} 1 & 1 & 2 & 1 & : & 5 \\ 2 & 3 & 1 & 2 & : & 2 \\ 4 & 5 & 3 & 0 & : & 7 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$= \begin{bmatrix} 1 & 1 & 2 & 1 & : & 5 \\ 0 & 1 & -5 & -4 & : & -8 \\ 0 & 1 & -5 & -4 & : & -13 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$= \begin{bmatrix} 1 & 1 & 2 & 1 & : & 5 \\ 0 & 1 & -5 & -4 & : & -8 \\ 0 & 0 & 0 & 0 & : & -5 \end{bmatrix}$$

$$\rho(A) = 2$$

$$\rho([A : B]) = 3$$

$\therefore \rho(A) \neq \rho([A : B]) \Rightarrow$ system is inconsistent and has no solution.

For what values of a and b the equations

$$\left. \begin{aligned} x + 2y + 3z &= 6 \\ x + 3y + 5z &= 9 \\ 2x + 5y + az &= b \end{aligned} \right\} \text{ have :-}$$

1. no solution
2. Unique solution
3. infinite solutions.

Ans. $[A : B] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 1 & 3 & 5 & 9 \\ 2 & 5 & a & b \end{array} \right]$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & a-6 & b-12 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$= \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & a-8 & b-15 \end{array} \right]$$

①. no solution

$$\rho(A) \neq \rho([A : B])$$

$$\therefore a-8=0$$

$$\boxed{a=8}$$

$$b-15 \neq 0$$

$$\boxed{b \neq 15}$$

$$\boxed{b = \text{any value}}$$

* ② unique solution

$$f(A) = f([A:B]) = 3$$

$b = \text{any value}$

$$\therefore a - 8 \neq 0$$

and

$$\boxed{a \neq 8}$$

and,

$$\boxed{\begin{matrix} b - 15 \neq 0 \\ b \neq -15 \end{matrix}}$$

③ infinite solution

$$f(A) = f([A:B]) < 3$$

$$\therefore a - 8 = 0$$

$$\boxed{a = 8}$$

and,

$$b - 15 = 0$$

$$\boxed{b = 15}$$

linearly dependent and independent vectors

Vectors $x_1, x_2, x_3, \dots, x_n$ are said to be linearly dependent if there exist scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ (not all zero) :

$$\lambda_1 x_1 + \lambda_2 x_2 + \dots + \lambda_n x_n = 0.$$

otherwise vectors are said to be linearly dependent.

Q. check whether the vectors :-

$$\begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \\ 2 \\ -4 \end{bmatrix}, \begin{bmatrix} -6 \\ 0 \\ -3 \\ -4 \end{bmatrix}$$

are linearly dependent or independent. If they are linearly dependent find the relation between them

Ans. let

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4 = 0$$

$$\lambda_1 \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 3 \\ 1 \\ 2 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 4 \\ 6 \\ 2 \\ -4 \end{bmatrix} + \lambda_4 \begin{bmatrix} -6 \\ 0 \\ -3 \\ -4 \end{bmatrix} = 0.$$

$$\therefore \begin{bmatrix} \lambda_1 + 3\lambda_2 + 4\lambda_3 - 6\lambda_4 \\ \lambda_2 + 6\lambda_3 \\ 2\lambda_1 + 2\lambda_2 + 2\lambda_3 - 3\lambda_4 \\ \lambda_1 + \lambda_2 - 4\lambda_3 - 4\lambda_4 \end{bmatrix} = 0$$

4x1.

$$\begin{aligned} \therefore \lambda_1 + 3\lambda_2 + 4\lambda_3 - 6\lambda_4 &= 0 \\ \lambda_2 + 6\lambda_3 &= 0 \\ 2\lambda_1 + 2\lambda_2 + 2\lambda_3 - 3\lambda_4 &= 0 \\ \lambda_1 + \lambda_2 - 4\lambda_3 - 4\lambda_4 &= 0 \end{aligned}$$

$$\therefore A = \begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 2 & 2 & 2 & -3 \\ 1 & 1 & -4 & -4 \end{bmatrix}$$

Note: $|A| \neq 0 \Rightarrow$ unique solution
 $\Rightarrow$ vectors linearly independent

$$|A| = 0 \Rightarrow$$

Now,

$$R_3 \rightarrow R_3 - 2R_1$$

$$R_4 \rightarrow R_4 - R_1$$

$$A = \begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & -4 & -6 & 9 \\ 0 & -2 & -8 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + 4R_2$$

$$R_4 \rightarrow R_4 + 2R_2$$

$$A = \begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 18 & 9 \\ 0 & 0 & 4 & 2 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - \frac{4}{18} R_3$$

$$A = \begin{bmatrix} 1 & 3 & 4 & -6 \\ 0 & 1 & 6 & 0 \\ 0 & 0 & 18 & 9 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\rho(A) = 3 < \text{number of variables}$

$\Rightarrow$ Non-trivial solution exists ~~and~~
~~system is~~ i.e. infinite solutions.

$\therefore$ equations:-

$$\lambda_1 + 3\lambda_2 + 4\lambda_3 - 6\lambda_4 = 0$$

$$\lambda_2 + 6\lambda_3 = 0$$

$$18\lambda_3 + 9\lambda_4 = 0$$

Put $\lambda_3 = t$.

$$\therefore \boxed{\lambda_4 = -2t}$$

$$\boxed{\lambda_1 = 2t}$$

$$\therefore \boxed{\lambda_2 = -6t}$$

$$\det t = 1 \quad \lambda_1 = 2 \quad \lambda_2 = -6 \quad \lambda_3 = 1 \quad \lambda_4 = -2$$

$$\therefore 2x_1 - 6x_2 + x_3 - 2x_4 = 0$$

$\Rightarrow$ Vectors are linearly dependent.

Imp Note:

$|A| = 0 \Rightarrow$ then vectors are linearly dependent.

$|A| \neq 0 \Rightarrow$ vectors are linearly independent

Imp

eg:- Find whether the vectors:-

$$\begin{bmatrix} 2 \\ 5 \\ 2 \\ -3 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ 5 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 14 \\ 14 \end{bmatrix}, \begin{bmatrix} 5 \\ 10 \\ 8 \\ 4 \end{bmatrix}$$

are linearly dependent or independent.

Ans.

$$A = \begin{matrix} & + & - & + & - \\ \begin{bmatrix} 2 & 3 & 4 & 5 \\ 5 & 6 & 5 & 10 \\ 2 & 5 & 14 & 8 \\ -3 & 2 & 14 & 4 \end{bmatrix} \end{matrix}$$

$$|A| = 2 \begin{vmatrix} 6 & 5 & 10 \\ 5 & 14 & 8 \\ 2 & 14 & 4 \end{vmatrix}$$

$$- 3 \begin{vmatrix} 5 & 5 & 10 \\ 2 & 14 & 8 \\ -3 & 14 & 4 \end{vmatrix}$$

$$+ 4 \begin{vmatrix} 5 & 6 & 10 \\ 2 & 5 & 8 \\ -3 & 2 & 4 \end{vmatrix} - 5 \begin{vmatrix} 5 & 6 & 5 \\ 2 & 5 & 14 \\ -3 & 2 & 14 \end{vmatrix}$$

$$\begin{aligned} |A| &= 2 [6(56-112) - 5(20-16) + 10(70-28)] \\ &\quad - 3 [5(56-112) - 5(8+24) + 10(28+42)] \\ &\quad + 4 [5(20-16) - 6(8+24) + 10(4+15)] \\ &\quad - 5 [5(70-28) - 6(28+42) + 5(4+15)] \end{aligned}$$

$$\begin{aligned} |A| &= 2 [336 - 20 + 420] - 3 [280 - 180 + 700] \\ &\quad + 4 [20 - 192 + 190] - 5 [210 - 420 + 95] \end{aligned}$$

$$= 2 [736] - 3 [800] + 4 [18] - 5 [305]$$

$$= 1472 - 2400 + 72 - 1525$$

$$\underline{|A| = -1383}$$

$$|A| \neq 0$$

$\therefore$ Vectors are linearly independent.

09/10/17

Characteristic EquationsEign Values & Eign Vectors.

For a square matrix A determinant of $|A - \lambda I| = 0$ is called characteristic equation of matrix A .

* Eign Values :-

The roots of the characteristic equation $|A - \lambda I| = 0$ are called eign values.

* Eign Vectors

The non-zero vector " X " is called Eign Vector corresponding to Eign value " λ " if :-

$$AX = \lambda X \quad [X \neq 0]$$

① For finding Eign vectors solve :-

$$(A - \lambda I)X = 0.$$

② If X is the Eign vector corresponding to Eign value " λ " then CX ($C \neq 0$) is also an Eign vector corresponding to same Eign value " λ ".

③ Number of independent solution are $(n-r)$

$$\boxed{\begin{matrix} \text{no. of variables} & - & \text{rank of matrix} \\ (n) & & (r) \end{matrix}}$$

eg:- $A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$

Find:-

- ① Characteristic equation
- ② Eigen values
- ③ Eigen vectors.

Ans. ① Characteristic equation

$$|A - \lambda I| = 0.$$

$$\left| \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0.$$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 3 & 3 & 4-\lambda \end{vmatrix} = 0.$$

$$(2-\lambda)((3-\lambda)(4-\lambda)-6) - 1(8-2\lambda-6) + 1(6-9+3\lambda) = 0.$$

$$(2-\lambda)(\cancel{12} + \lambda^2 - 7\lambda + 6) - (2-2\lambda) + 1(3\lambda-3) = 0.$$

$$(2-\lambda)(\lambda^2 - 7\lambda + 6) - 2 + 2\lambda + 3\lambda - 3 = 0.$$

$$2\lambda^2 - 14\lambda + 12 - \lambda^3 + 7\lambda^2 - 6\lambda - 5 + 5\lambda = 0.$$

$$\lambda^3 - 9\lambda^2 + 15\lambda - 7 = 0.$$

② Eigen values.

$$\lambda^3 - 9\lambda^2 + 15\lambda - 7 = 0.$$

$$\lambda^2(\lambda-1) - 8\lambda(\lambda-1) + 7(\lambda-1) = 0.$$

$$(\lambda-1)(\lambda^2 - 8\lambda + 7) = 0.$$

$$(\lambda-1)(\lambda^2 - 7\lambda - \lambda + 7) = 0.$$

$$(\lambda-1)(\lambda(\lambda-7) - 1(\lambda-7)) = 0$$

$$(\lambda-1)(\lambda-1)(\lambda-7) = 0$$

$$\lambda = 1, 1, 7$$

③ Eigen Vectors

(a) corresponding to $\lambda = 7$

$$(A - \lambda I)X = 0.$$

$$\left(\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix} - \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix} \right) X = 0.$$

$$\begin{bmatrix} -5 & 1 & 1 \\ 2 & -4 & 2 \\ 3 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

$\underbrace{\hspace{1.5cm}}_{3 \times 3} \quad \underbrace{\hspace{1.5cm}}_{3 \times 1}$

Now, $A = \begin{bmatrix} -5 & 1 & 1 \\ 2 & -4 & 2 \\ 3 & 3 & -3 \end{bmatrix}$

$$R_1 \rightarrow R_1 + 3R_2$$

$$\begin{bmatrix} 1 & -11 & 7 \\ 2 & -4 & 2 \\ 3 & 3 & -3 \end{bmatrix}$$

~~$R_2 \rightarrow 3R_2$~~ $R_2 \rightarrow R_2 - 2R_1$
 $R_3 \rightarrow R_3 - 3R_1$

$$A = \begin{bmatrix} 1 & -11 & 7 \\ 0 & 18 & -12 \\ 0 & 36 & -24 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - 2R_2$$

$$A = \begin{bmatrix} 1 & -11 & 7 \\ 0 & 18 & -12 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\rho(A) = 2}$$

∴ Number of independent solutions

$$n - \rho.$$

$$3 - 2 = 1.$$

∴ Equations :-

$$x_1 - 11x_2 + 7x_3 = 0$$

$$18x_2 - 12x_3 = 0.$$

~~for~~, Put $x_3 = t$

$$x_1 = \frac{t}{3}$$

$$x_2 = \frac{2t}{3}$$

$$\Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t/3 \\ 2t/3 \\ t \end{bmatrix}$$

Put $t=3$

$$X = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

is an eigen vector

$$\underline{X = \hat{i} + 2\hat{j} + 3\hat{k}}$$

(b) corresponding to $\lambda = 1$.

$$(A - \lambda I) = 0$$

$$\left| \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right| X = 0$$

$$\left| \begin{array}{ccc|c} 1 & 1 & 1 & \lambda \\ 2 & 2 & 2 & \\ 3 & 3 & 3 & \end{array} \right| = 0$$

Now, $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 - 3R_1$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 1$$

No. of independent solutions $= n - \rho$
 $= 3 - 1 = 2$.

equation :-

$$x_1 + x_2 + x_3 = 0$$

$$x_2 = t_1 \quad x_3 = t_2 \quad x_1 = -t_1 - t_2$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -t_1 - t_2 \\ t_1 \\ t_2 \end{bmatrix}$$

Put $t_1 = 1 \quad t_2 = 0$
 and $t_2 = 1 \quad t_1 = 0$.

$$X = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

are eigen vectors corresponding to $\lambda = 1$

$$8 + 20 - 34 + 21$$

$$6 - 4$$

$$27$$

Q.

$$A = \begin{bmatrix} -2 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 6 \end{bmatrix}$$

Find :-

- ① characteristic equation
- ② eigen values.
- ③ eigen vectors

Ans.

① characteristic equation

$$|A - \lambda I| = 0.$$

$$\left| \begin{bmatrix} -2 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 2 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0.$$

$$\begin{bmatrix} -2-\lambda & 2 & 1 \\ 2 & 1-\lambda & 2 \\ 1 & 2 & 6-\lambda \end{bmatrix} = 0.$$

$$(-2-\lambda)(1-\lambda)(6-\lambda) - 4 - 2(12-2\lambda-2) + 1(4-1+\lambda) = 0.$$

$$(-2-\lambda)(6+\lambda^2-7\lambda-4) - 20 + 4\lambda + 3 + \lambda = 0$$

$$-2\lambda^2 - \lambda^3 + 14\lambda + 7\lambda^2 - 4 - 20 + 4\lambda + 3 + \lambda = 0.$$

$$-\lambda^3 + 5\lambda^2 + 17\lambda - 21 = 0.$$

$$\lambda^3 - 5\lambda^2 - 17\lambda + 21 = 0.$$

② Eigen values

$$\lambda^2(\lambda-1) - 4\lambda(\lambda-1) + 21(\lambda-1) = 0.$$

$$(\lambda^2 - 4\lambda + 21)(\lambda-1) = 0.$$

$$(\lambda+1)(\lambda^2-7)$$

$$(\lambda-1)(\lambda+3)(\lambda-7)=0$$

$$\lambda = 1, -3, 7$$

③ Eign ~~value~~ vectors

(a) corresponding to 1.

$$(A - \lambda I) x = 0.$$

$$\left[\begin{array}{ccc|ccc} -2 & 2 & 1 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 1 & 2 & 6 & 0 & 0 & 1 \end{array} \right] x = 0.$$

$$\begin{pmatrix} -3 & 2 & 1 \\ 2 & 0 & 2 \\ 1 & 2 & 5 \end{pmatrix} x = 0.$$

Now, $A = \begin{bmatrix} -3 & 2 & 1 \\ 2 & 0 & 2 \\ 1 & 2 & 5 \end{bmatrix}$

$$R_1 \leftrightarrow R_3$$

$$= \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 2 \\ -3 & 2 & 1 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$R_3 \rightarrow R_3 + 3R_1$$

$$A \approx \begin{bmatrix} 1 & 2 & 5 \\ 0 & -4 & -8 \\ 0 & 8 & 16 \end{bmatrix}$$

$$R_2 \rightarrow R_3 + 2R_2$$

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 0 & -4 & -8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\rho(A) = 2}$$

$$\begin{aligned} \text{number of variables} &= n - \rho \\ &= 3 - 2 = 1. \end{aligned}$$

equations :-

$$\begin{aligned} x_1 + 2x_2 + 5x_3 &= 0 \\ -4x_2 - 8x_3 &= 0 \end{aligned}$$

Put $x_3 = t$.

$$x_1 = t$$

$$x_2 = -2t$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ -2t \\ t \end{bmatrix}$$

10/10/19

Q1. Find the values of a and b such that rank of matrix : $A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 2 & 1 & -1 & 2 \\ 6 & -2 & a & b \end{bmatrix}$ is 2.

Q2. Are the vectors $x_1 = [1 \ 3 \ 4 \ 2]$

$$x_2 = [3 \ -5 \ 2 \ 2]$$

$$x_3 = [2 \ -1 \ 3 \ 2]$$

L.D, if dependent, find relation.

Q3. For what values of a, b the system:

$$x + 2y + 3z = 6.$$

$$x + 3y + 5z = 9$$

$$2x + 5y + az = b.$$

(i) no solution

(ii) Unique

(iii) infinitely many solutions.

Q4. Find eigen value and eigen vector of:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Q5. State and verify Cayley-Hamilton Theorem for $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ and use it to find A^{-1} and A .

$$\begin{bmatrix} \lambda_1 + 3\lambda_2 + 2\lambda_3 \\ 3\lambda_1 - 5\lambda_2 - \lambda_3 \\ 4\lambda_1 + 2\lambda_2 + 3\lambda_3 \\ 2\lambda_1 + 2\lambda_2 + 2\lambda_3 \end{bmatrix} = 0$$

$$\lambda_1 + 3\lambda_2 + 2\lambda_3 = 0$$

$$3\lambda_1 - 5\lambda_2 - \lambda_3 = 0$$

$$4\lambda_1 + 2\lambda_2 + 3\lambda_3 = 0$$

$$2\lambda_1 + 2\lambda_2 + 2\lambda_3 = 0$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 3 & -5 & -1 \\ 4 & 2 & 3 \\ 2 & 2 & 2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$R_3 \rightarrow R_3 - 4R_1$$

$$R_4 \rightarrow R_4 - 2R_1$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -14 & -7 \\ 0 & -10 & -5 \\ 0 & -4 & -2 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_4$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -10 & -5 \\ 0 & -10 & -5 \\ 0 & -4 & -2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -10 & -5 \\ 0 & 0 & 0 \\ 0 & -4 & -2 \end{bmatrix}$$

$$R_1 \rightarrow 10R_1 - 4R_2$$

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & -10 & -5 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho(A) = 2$$

Cayley - Hamilton Theorem.

Every square matrix satisfies its characteristic equation.

eg:- For the matrix $\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$. ~~Verify~~ verify Cayley Hamilton Theorem.

Ans Characteristic equation is:-

$$|A - \lambda I| = 0.$$

$$\left| \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0.$$

On solving:-

$$-\lambda^3 + 5\lambda^2 - 7\lambda + 3 = 0.$$

To verify Cayley Hamilton Theorem.

ie to prove :- $-A^3 + 5A^2 - 7A + 3I = 0$.

$$\text{LHS} = -A^3 + 5A^2 - 7A + 3I.$$

$$= - \begin{bmatrix} 14 & 13 & 13 \\ 0 & 1 & 0 \\ 13 & 13 & 14 \end{bmatrix} + 5 \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} - 7 \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -14 & -13 & -13 \\ 0 & -1 & 0 \\ -13 & -13 & -14 \end{bmatrix} + \begin{bmatrix} 25 & 20 & 20 \\ 0 & 5 & 0 \\ 20 & 20 & 25 \end{bmatrix} = \begin{bmatrix} 14 & 7 & 7 \\ 0 & 7 & 0 \\ 7 & 7 & 14 \end{bmatrix} + \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= 0$$

$$= \text{RHS.}$$

hence proved.

→ Reduction of a matrix to Diagonal Matrix

Let a square matrix A of order " n " has " n " linearly independent ~~vectors~~ eigen vectors, then we can find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

matrix

Let A be of order = 3 having eigen value $\lambda_1, \lambda_2, \lambda_3$ and corresponding eigen vectors are ~~x_1, y_1, z_1~~ $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix}, \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}, \begin{pmatrix} x_3 \\ y_3 \\ z_3 \end{pmatrix}$

respectively. Then,

$$P = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$D = P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

Note: P is called modal matrix and D is called spectral matrix of A .

Remark: -

① If $D = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$, then $D^n = \begin{bmatrix} a^n & 0 & 0 \\ 0 & b^n & 0 \\ 0 & 0 & c^n \end{bmatrix}$

② If $D = P^{-1}AP$, then $A = PDP^{-1}$

$\Rightarrow$

$$A^2 = (PDP^{-1})(PDP^{-1})$$

$$A^2 = PD^2P^{-1}$$

similarly,

$$A^n = PD^nP^{-1}$$

∴ Find the matrix P which transform the matrix $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ to the diagonal matrix and then find the diagonal matrix and hence find A^4 .

Ans. (1) characteristic equation

(2) Eigen values

$$\lambda = -2, 3, 6$$

(3) Eigen vectors :-

$$(-1, 0, 1); (1, -1, 1); (1, 2, 1)$$

$$\therefore P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{cc} m_{11} & m_{21} \\ m_{12} & m_{22} \\ m_{13} & \end{array} \left[\begin{array}{ccc|ccc} -1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_1$$

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 1 & 0 \\ 0 & 2 & 2 & 1 & 0 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 + 2R_2$$

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 1 & 1 & 0 & 0 \\ 0 & -1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 6 & 1 & 2 & 1 \end{array} \right]$$

$$\therefore P^{-1} = \frac{1}{6} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$D = P^{-1}AP$$

$$D = \frac{1}{6} \begin{bmatrix} -2 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

$$A^4 = P D^4 P^{-1}$$

~~$$= \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 16 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 1296 \end{bmatrix} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$~~

$$= \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 16 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 1296 \end{bmatrix} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} -16 & 81 & 1296 \\ 0 & -81 & 2592 \\ 16 & 81 & 1296 \end{bmatrix} \begin{bmatrix} -3 & 0 & 3 \\ 2 & -2 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

~~$$= \frac{1506}{5436} \begin{bmatrix} 251 & 405 & 235 \\ 405 & 891 & 405 \\ 235 & 405 & 251 \end{bmatrix}$$~~

$$A^4 = \begin{bmatrix} 251 & 405 & 235 \\ 405 & 891 & 405 \\ 235 & 405 & 251 \end{bmatrix}$$

Similar Matrices:-

Two matrices A and B are said to be similar if there exist a non-singular matrix P such that :-

$$B = P^{-1}AP$$

∴ Diagonal matrix is similar to matrix A .

→ Properties of similar matrices:-

① $|A| = |B|$

same

② Similar matrices have characteristic equations

→ Properties of eigen values & vectors:-

1. If x is eigen vector corresponding to eigen value λ then cx is also eigen vector corresponding to same eigen value λ .

2. If λ is the eigen value of matrix A :-

(a) λ is also eigen value of A^T

(b) $1/\lambda$ is eigen value of A^{-1}

(c) λ^m is the eigen value of A^m

3. The eigen values of a triangular matrix (Diagonal matrix) are just the diagonal elements of the matrix.

4. The eigen values of an ~~id~~ idempotent matrix $(A^2 = A)$ are ~~any~~ either zero or unity.

5. The eigen values of a hermitian matrix ^(complex) are real.
6. The eigen values of a skew hermitian matrix are purely imaginary or zero.
7. The eigen values of a unitary matrix ^($AA^T = I$) has absolute value 1.
8. The eigen values of an orthogonal matrix ^($AA^T = I$) has absolute value 1.
9. The sum of eigen values of a matrix is equal to sum of elements of its principal diagonal and it is called Trace of a matrix.
10. The product of eigen values of a matrix is equal to its determinant.

eg:- The eigen values of matrix A are $(2, -3, 7)$.
 Find eigen values of A^2 and A^{-1} .
 Also find trace of matrix A and determinant of A.

Ans. • eigen values of A^2
 $(2^2, (-3)^2, (7)^2)$

$$= (4, 9, 49)$$

• eigen values of A^{-1}
 $(2^{-1}, (-3)^{-1}, 7^{-1})$

$$= \left(\frac{1}{2}, -\frac{1}{3}, \frac{1}{7} \right)$$

- Trace of $A = 2 - 3 + 7$
 $= 6$

- $|A| = 2 \times (-3) \times 7$
 $= -42$

13/10/17

→ COMPLEX MATRICES :-

* Hermitian Matrix.

- A matrix A is called hermitian if :-

$$A^* = (\overline{A})^T = A$$

- The leading diagonal elements of a hermitian matrix are all real.

* Skew Hermitian Matrix

- A matrix A is called skew hermitian if

$$A^* = -A$$

- All diagonal elements are purely zero or imaginary

* Unitary matrix

A square matrix A is called unitary if :-

$$AA^* = A^*A = I$$

g:- Show that the matrix :-

$$U = \frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$$

is a unitary matrix.

$$U^* = \frac{1}{2} \begin{bmatrix} 1-i & -1-i \\ 1-i & 1+i \end{bmatrix}^T$$

$$U^* = \frac{1}{2} \begin{bmatrix} 1-i & 1-i \\ -1-i & 1+i \end{bmatrix}$$

$$UU^* = \frac{1}{4} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix} \begin{bmatrix} 1-i & 1-i \\ -1-i & 1+i \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 2+2 & 2-2 \\ \cancel{2-2} & 2+2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$UU^* = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U^*U = \begin{bmatrix} 1-i & 1-i \\ -1-i & 1+i \end{bmatrix} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore \underline{U^*U = UU^* = I}$$

eg:- If A and B are complex matrices then show that :-

$$(i) (A^*)^* = A$$

$$(ii) (A+B)^* = A^* + B^*$$

$$(iii) (AB)^* = B^* A^*$$

$$(iv) (kA)^* = \bar{k} A^*$$

$$(v) B^* A B = \text{Hermitian or skew hermitian if } A \text{ is hermitian or skew hermitian.}$$

(vi) If A is unitary matrix then A^T is unitary matrix.

(vii) If A is unitary matrix then A^{-1} is also unitary matrix.

(viii) If A and B are unitary matrix then AB is also unitary matrix.

Note:

① Every square matrix A can be expressed as $R + iS$, where both R and S are hermitian matrices.

② Every square matrix A can be written as sum of hermitian and skew hermitian matrix.

$$① \Rightarrow R = \frac{A + A^*}{2}$$

$$S = \frac{A - A^*}{2i}$$

$$② \Rightarrow R = \frac{A + A^*}{2}, \quad S = \frac{A - A^*}{2i}$$

eg:-

$$A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix}$$

Express as sum
of Hermitian
and skew hermitian
matrices.

$$\bar{A}' = \begin{bmatrix} 1-i & 2 & 5+5i \\ -2i & 2-i & 4-2i \\ -1-i & -4 & 7 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 1-i & -2i & -1-i \\ 2 & 2-i & -4 \\ 5+5i & 4-2i & 7 \end{bmatrix}$$

$$R = \frac{A + A^*}{2}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 2-2i & 4-6i \\ 2+2i & 4 & 2i \\ 4+6i & -2i & 14 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 1-i & 2-3i \\ 1+i & 2 & i \\ 2+3i & -i & 7 \end{bmatrix}$$

$$S = \frac{A - A^*}{2}$$

$$= \frac{1}{2} \begin{bmatrix} 2i & 2+2i & 6-4i \\ 2i-2 & 2i & 8+2i \\ -6-4i & -8+2i & 0 \end{bmatrix}$$

$$= \begin{bmatrix} i & 1+i & 3-2i \\ i-2 & i & 4+i \\ -3-2i & -4+i & 0 \end{bmatrix}$$

15/10/17 QUADRATIC FORM

A homogenous polynomial of 2nd degree in any number of variables is called a quadratic form.

eg:- $ax^2 + 2hxy + by^2$
 $ax^2 + by^2 + cz^2 + 2hxy + 2gyz + 2fzx$

→ Quadratic form in Matrix Form.

* $ax^2 + 2hxy + by^2$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & h \\ h & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = X^T A X.$$

* $ax^2 + by^2 + cz^2 + 2hxy + 2gyz + 2fzx$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{matrix} x & y & z \\ \begin{bmatrix} a & h & f \\ h & b & g \\ f & g & c \end{bmatrix} \end{matrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

eg: Find matrix A of quadratic form.

* $x_1^2 + 4x_2^2 + 5x_3^2 + 2x_1x_2 + x_2x_3 + 3x_1x_3$

Ans. $\begin{bmatrix} x_1 & x_2 & x_3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3/2 \\ 2 & 4 & 1/2 \\ 3/2 & 1/2 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

Ques

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{bmatrix}$$

find quadratic form

Ans.

$$\text{quadratic form} = x^T A x$$

$$\begin{bmatrix} x & y & z \end{bmatrix} \begin{matrix} x & y & z \\ 1 & 2 & 5 \\ 2 & 0 & 3 \\ 5 & 3 & 4 \end{matrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$x^2 + 4z^2 + \underline{10xy} + 10xz + 6yz$$

→ CANONICAL FORM

If the quadratic form $x^T A x$ can be reduced to the form:- ~~$\sum_{i=1}^n \lambda_i x_i^2$~~ ~~$\sum_{i=1}^n \lambda_i y_i^2$~~

$$\sum_{i=1}^n \lambda_i y_i^2 = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2$$

then $y^T B y$ is called canonical form of given quadratic form.

→ (EP) INDEX and SIGNATURE of Quadratic Form

The number of +ve terms in canonical form is called index of quadratic form.

The number of +ve terms minus number of -ve terms is called signature of quadratic form.

→ Types of Quadratic Form.

Quadratic form is called
(i) the definite if rank of matrix $A = n$ and
index $= n$ i.e. $\rho(A) = P = n$.

(ii) -ve definite if $\rho(A) = n$ & $P = 0$.

(iii) the semi-definite if $\rho(A) < n$ &
 $P = \rho(A)$

(iv) -ve semi-definite if $\rho(A) < n$ &
 $P = 0$.

(v) if canonical form has both +ve and -ve
terms then quadratic form is called
indefinite

Note: Orthogonal Vectors :-

Vectors x and y are said to be orthogonal
if $\boxed{xy^T = 0}$

→ Reduction of quadratic form to canonical form

convert :-

$$8x^2 + 7y^2 + 3z^2 - 12xy + 4xz - 8yz$$

into canonical form. What is rank of A.

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

eigen values are:- $0, 3, 15$

eigen vectors are:- $(1, 2, 2)$ $(2, 1, -2)$ $(2, -2, 1)$

These vectors are pair-wise orthogonal
i.e. $x_1 \cdot x_2^T = 0 = x_2 \cdot x_3^T = x_3 \cdot x_1^T$

Normalised Model matrix.

$$P = \left[\frac{x_1}{\|x_1\|}, \frac{x_2}{\|x_2\|}, \frac{x_3}{\|x_3\|} \right]$$

$$\|x_1\| = \sqrt{1+4+4} = 3$$

$$\|x_2\| = \sqrt{4+1+4} = 3$$

$$\|x_3\| = \sqrt{4+4+1} = 3$$

$$P = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

Property of orthogonal matrix $PP^T = I$.

$$P^{-1} = P^T = \begin{bmatrix} 1/3 & 2/3 & 2/3 \\ 2/3 & 1/3 & -2/3 \\ 2/3 & -2/3 & 1/3 \end{bmatrix}$$

$$D = P^T A P = P^T A P$$

$$D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$

$\therefore$ Canonical form.

$$y^T A y = \begin{bmatrix} y_1 & y_2 & y_3 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$= 3y_2^2 + 15y_3^2$$

$$\therefore r(A) = 2$$

$$P = 2$$

$$\text{signature} = 2$$

and A is true semi-definite.

convert

eg:- $6x_1^2 + 3x_2^2 + 3x_3^2 - 4x_1x_2 - 2x_2x_3 + 4x_3x_1$

into canonical form.

Ans.

$$A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

characteristic equation :-

$$\lambda^3 - 12\lambda^2 + 36\lambda - 32 = 0$$